# Zaw Wana


Platform / Site Reliability Engineer — Kubernetes | Linux | Observability

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## PROFESSIONAL SUMMARY


Certified Kubernetes Administrator running production container platforms and Linux fleets in enterprise and public-sector environments. Day-2 operations across the full cluster lifecycle: node lifecycle and etcd health, RBAC and SCC policy, network policies, routes and HAProxy ingress, resource quotas, and persistent storage. First-responder experience tracing failures from node to pod to service, with root-cause analysis fed back into runbooks, and ELK/Prometheus observability to catch failures before users do. Also owns the Linux OS layer beneath the platform — patching cycles, kernel and resource tuning, hardening, and storage — plus highly available data tiers and secrets through Vault and CyberArk. Currently supporting a government AI infrastructure programme under IM8-aligned security controls. Computing Science honours graduate, Singapore Institute of Technology.


## CORE TECHNICAL SKILLS

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|---|---|
| **Kubernetes & Containers** | Kubernetes (CKA) — node lifecycle, etcd health, RBAC, network policies, resource quotas, PV/PVC & storage classes; OpenShift Container Platform (OCP 4.x) projects, Routes, SCCs, HAProxy ingress routing and timeout tuning, private registry & image mirroring; K3s, kubeadm, Cilium, MetalLB, Helm, Docker |
| **Reliability & Observability** | Incident response and root-cause analysis, runbooks, ELK / Elasticsearch / Logstash / Kibana, OpenSearch, Prometheus, Grafana, Zabbix, log aggregation, alerting, capacity & performance monitoring |
| **High Availability** | MariaDB Galera (multi-master HA), HAProxy proxy tier for connection routing and health checking, backup & recovery, JVM/Java workload tuning (heap, Metaspace, GC flags, container memory budgeting) |
| **Linux Platform** | RHEL-family (RHEL/CentOS/Rocky) & Ubuntu administration, OS patching & kernel upgrades, reboot sequencing, dnf/yum package management, systemd, SELinux, firewalld, auditd, ulimits/cgroups, NFS & iSCSI/SMB storage mounts, LVM, Windows Server |
| **Automation & IaC** | Terraform, Ansible, Jenkins, GitLab CI, GitHub Actions, ArgoCD / GitOps, Git, Python, Bash, operational runbooks |
| **Security & Hardening** | CIS benchmark hardening, IM8-aligned controls, vulnerability remediation & CVE triage, SSH cipher/MAC hardening, HashiCorp Vault, CyberArk PAM, LDAP, Azure AD (Entra ID) IAM & RBAC, IAM, secrets management, IPSec VPN, DevSecOps |
| **Data & Cloud** | MySQL, MongoDB, Redis; AWS (EC2, S3, IAM, CloudWatch, VPC), Azure (Entra ID, RBAC, Policy), hybrid & on-prem integration, Proxmox virtualisation |

## WORK EXPERIENCE

### DevSecOps / Platform Engineer — NCS — Nov 2025 — Present

*Video intelligence and enterprise AI infrastructure platforms — on-prem estate, government security baseline.*

- Operate the OpenShift Container Platform estate the workloads run on — projects and namespaces, RBAC and SCC policy, network policies and resource quotas, Routes and HAProxy ingress, and persistent storage via PV/PVC and storage classes
- Built and operate MariaDB on a Galera cluster — synchronous multi-master replication so the database survives node loss without failover downtime or manual promotion; front-ended by a proxy tier (HAProxy) for connection routing and health checking
- Operate ELK-based monitoring across cluster, systemd service, and MariaDB logs — aggregated into dashboards and alerting so failures surface before users report them
- Own the Linux OS layer beneath the platform — patching cycles, kernel and package management, resource limits, and storage mounts — sequenced so production services stay up through the change window
- Day-1 build and Day-2 operations on an enterprise AI infrastructure platform, plus Database Engineer ownership of schema design, tuning, and backup and recovery
- Provision platform infrastructure with Terraform so environments are described in code and rebuilt from it rather than reassembled by hand

- Apply and validate OS hardening against CIS-style benchmarks and government (IM8) security requirements; remediate vulnerability-scan findings and produce the supporting remediation evidence
- Manage privileged access with CyberArk and centralise service credentials in HashiCorp Vault, removing long-lived secrets from configuration files and node images
- Administer identity and access governance on Azure AD (Entra ID) — user/group lifecycle, RBAC role assignments, and Azure Policy enforcement — alongside on-prem LDAP
- Run LDAP-backed authentication and group authorisation across a mixed RHEL-family Linux and Windows Server estate
- Configure IPSec VPN tunnels so field mobile devices reach internal systems without exposing them publicly

**DevOps / Infrastructure Engineer — Singtel** **Sep 2024 — Jul 2025**

*Infrastructure, delivery, and operations behind Singtel's NaaS, 5G, and multi-access edge computing (MEC) products.*

- Ran day-to-day operations across Linux servers and Kubernetes clusters as first responder, tracing failures from node to pod to service
- Carried out root-cause analysis on those incidents and fed the findings back into runbooks
- Administered Kubernetes clusters supporting edge workloads — node management, workload placement, and resource limits
- Standardised application deployments across those clusters by packaging them with Helm
- Implemented full-stack Zabbix monitoring across infrastructure, network, and application metrics
- Maintained and extended the Jenkins CI/CD pipelines behind application delivery
- Delivered application rollouts through ArgoCD with Git as the source of truth, so cluster drift was reconciled automatically instead of patched in place
- Troubleshot and performance-tuned OpenSearch, MySQL, Redis, and MongoDB
- Troubleshot and performance-tuned containerised application workloads
- Owned OS patching and update cycles across the Linux server fleet — keeping systems current and vulnerabilities closed without taking production services down
- Automated infrastructure provisioning with Terraform
- Automated daily Docker and virtualised operations with Ansible
- Wrote Python automation for security tooling, including IoC IP filtering
- Wrote Python automation for operational tasks, including system health checks

## PLATFORM ENGINEERING — SELF-DIRECTED

- Two-tier homelab Kubernetes estate, continuously operated: a 3-node kubeadm cluster (Cilium CNI, MetalLB L2 load balancing, containerd) running production workloads, plus a K3s cluster for development — both run through real upgrade, certificate-rotation, and failure scenarios
- GitOps delivery with ArgoCD in an app-of-apps pattern — a root Application bootstraps every other workload from Git, with automated sync, pruning, and self-heal so cluster drift is reconciled rather than patched in place
- Hosts a personal portfolio site and internal homelab services on the platform, with ingress and TLS termination, persistent storage classes, RBAC and namespace quotas, a private image registry, and a full Prometheus/Grafana + ELK observability stack — built to enterprise patterns
- HA data tier: MariaDB Galera cluster behind an evaluated proxy layer (HAProxy / MaxScale / ProxySQL), with backup, restore, and node-loss drills
- Platform roadmap in progress: Gateway API for HTTP routing in place of classic ingress, and Trivy image and manifest scanning wired into the delivery pipeline
- Bash-based CKA exam simulator running against a disposable k3d sandbox; AWS serverless projects for SAA preparation

## CERTIFICATIONS & EDUCATION

**Certified Kubernetes Administrator (CKA) — Cloud Native Computing Foundation · ID LF-ijwzjz1ojo** **Aug 2026**

**BSc Computing Science (Honours) — Singapore Institute of Technology** **Aug 2022 — Apr 2025**

**Diploma in Business Process with Systems Engineering — Temasek Polytechnic** **Apr 2017 — Feb 2020**